

Effects of the COVID-19 Pandemic on Park Visitation Measured by Social Media

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ABSTRACT

The COVID-19 pandemic and the resulting economic recession have negatively affected many people's physical, social, and psychological health. Parks and green-spaces may have ameliorated the negative effects of the pandemic by creating opportunities for outdoor recreation and nature exposure, while other public activities and gatherings were restricted due to risk of disease transmissions. Using park visitation estimates derived from 140K Instagram images shared in four US metropolitan areas, this study investigates trends in park use over the span of the COVID-19 pandemic. We find that while COVID-related stay-at-home orders are associated with shifts in park visitation, the cities that we analysed do not follow uniform trends. Our analyses suggest that future research may be able to explain variability in park visitation based on local factors such as park location and the socio-demographics of visitors. However, the research community does not currently have access to the volume and resolution of data that is necessary to study the issue. There is an urgent need for the CSCW community and social computing researchers to address this data gap if we are to understand the impacts of the pandemic, plan for urban resiliency, and ensure equitable access to parks and other shared resources.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in collaborative and social computing.**

KEYWORDS

nature exposure, outdoor recreation, social media analysis, park visitation, COVID-19

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1 INTRODUCTION

Urban parks are green infrastructure that provide people with opportunities for outdoor recreation [12], build community [20], and improve the health and well-being of residents [13, 16, 18, 19]. During the COVID-19 pandemic, indoor social gatherings and other recreational activities carried serious risk of disease transmission in most communities and therefore many public locations were forced to close as a safety precaution. Meanwhile, parks and green-spaces provided comparatively low-risk environments, and anecdotal evidence indicated that park use increased in 2020 [3, 8, 14, 17].

Several recent studies on the impact of the pandemic on park visitation have reached differing conclusions. A global analysis of park visitation patterns using data collected primarily from Google's Community Mobility Reports [2] showed that park visitation has increased since February 2020, compared to visitor numbers prior to the COVID-19 pandemic [8]. Other studies, such as [11] showed a 14% decline in urban park visitation during the pandemic, based on a study of US urban parks using data from SafeGraph [9]. In [8] authors conducted a longitudinal study of Instagram posts from 100K users (1,185 green spaces) in four Asian cities and showed a 5.3% surge in during the pandemic.

Existing studies of park recreation during the COVID-19 pandemic have several limitations which may help to explain their contrasting conclusions. Mobility data, such as those provided by Google, capture only a subset of mobile phone users: those who have intentionally turned on the Location History setting of their device. Since these data have never been validated through direct comparison with on-site estimates of park use, it is unclear whether they are an accurate proxy for visits, or if mobility data require adjustments. Studies relying on social media such as Instagram to estimate park visitation may be similarly biased towards only a subset of visitors [22]. There is evidence that the online popularity of urban parks on social media platforms is correlated with actual park visits [7, 10], although the relationship varies by social media platform [21, 23]. Additionally, few studies have analysed trends in park visitation over time with regards to the local stay-at-home order (SAHO) guidelines and park closures that were in effect at different times in different locations. Nor have there been many comparisons of differences among parks — within and between cities — and factors that may be driving variability in space. Google mobility data do not allow for such park level comparisons because they are aggregated by large geographies such as states or counties. This prevents the fine-grain analysis that is often needed by practitioners who manage parks.

In this paper, we seek to improve our understanding of how the COVID-19 pandemic is impacting park visitation by performing a

contextualized longitudinal study of the ten largest parks in four cities in the U.S. Our analyses quantify nature visits using the number of images that were shared by visitors of the same parks on Instagram. In order to study whether socio-geographical factors can explain the context dependency of park visitations, we investigate differences between parks at the urban fringe compared to core urban neighborhood parks where residents might seek sanctuary. We also ask whether parks in areas with more vulnerable populations receive less visitation before and during the pandemic. Our results show that while there are large shifts in park visitation patterns coincident with the pandemic, the cities that we analysed do not exhibit the same visitation trends. Some of the difference that we observed are due to the setting of the park, revealed by comparing urban and urban-fringe parks. Finally, our analyses highlight that while there is tremendous value in social media and mobility data, the research community does not have access the volume and resolution of data that it necessary to fully understand the issues.

2 METHODOLOGY

2.1 Site Selection

Four cities were chosen for this study by selecting the two most populated US metropolitan cities from the West (Los Angeles and Phoenix) and two from the East (Chicago and New York City). Within each city, we selected the ten largest parks. We also ensured that every park was designated as local park according to the PAD-US standardized designations [4], and none were primarily sports facilities. For New York City, Central Park was removed from the study because its collection of Instagram posts was very large, even in comparison to other cities.

2.2 Data Collection

We compiled metadata about all images shared publicly on Instagram and assigned to a location within the selected parks between Jan 2019–Jan 2021 [as in 23]. Since Instagram no longer provides an automated method to query locations, we first manually searched for locations in the Instagram web interface, using the park name and major features as search terms. Then we used Instagram's *graphql* endpoint to collect information about images tagged to each location. In selecting the location IDs we eliminated those that corresponded purely to the private Points of Interests within a park (e.g., a zoo or coffee shop). This helped ensure that pre- and post-covid are comparable and tied to use of the green spaces instead of businesses within the park. The chosen parks across all cities contained approximately over 140,000 unique posts from Instagram. In order to compare results per city, we compiled information on the timing of local declarations and mandates including the first emergency declaration, SAHO, and public mask mandate. The per-city information was obtained from a data repository maintained by [5, 6]. We categorized each park as urban or urban-fringe depending on its distance from the city center. We used the Social Vulnerability Index (SVI) as published [1] to contextualize parks in terms of the socio-economic vulnerability of the surrounding neighborhoods. To assign the SVI of parks that spanned multiple census tracts, we selected the SVI corresponding with the centre of the park.

2.3 Metrics

We measured the total numbers of unique visitors per day who posted geo-located photographs from a particular park over a given time period — also termed “photo-user-days” (*PUD*) by ([22]). Parks have seasonal trends in visitation, and these trends differ according to the climate of each city. To control for seasonality, we measured monthly changes in PUD from 2019 to 2020. We also controlled for year-to-year changes in park visitation and the popularity of parks on Instagram (for reasons unrelated the pandemic). Changes in monthly PUD after the date of SAHOs were compared to year-to-year changes observed during Jan and Feb 2020, prior to the pandemic.

3 ANALYSIS

Our longitudinal analyses captured changes in visits to the selected parks. In Los Angeles and Phoenix, visitation decreased after the SAHO was enacted in April 2020 (Figure 1). Chicago parks exhibited a smaller decrease in visitation after the SAHO, although this outcome could be due to the anomalously high visitation in August 2019 when a festival occurred in one park. In New York, park visitation increased after the SAHO, exceeding the peak visitation values in 2019. These results indicate that trends in park visitation were not consistent among cities, even with regards to specific COVID policies.

There were pre-SAHO changes in year-to-year park visitation that were evident by comparing Jan–Mar 2019 versus the same months in 2020 (Figure 2). These differences represent the background year-to-year difference in park visitation due to changes in the popularity of parks and social media from 2019 to 2020. The discrepancy between pre- and post-SAHO changes provides insights into how park visitation was impacted by the pandemic or other factors outside of park and social media popularity. Generally, once a SAHO was enacted, the percentage change in park visitation was lower than the change over the two months prior to the SAHO (Figure 2). In both Los Angeles and New York, visitation did not return to pre-SAHO levels. On the other hand, park use in Chicago did return to pre-SAHO levels of increasing visitation nine months after the SAHO. This result can be seen in with the first red square symbol at 26% above the baseline in Figure 2. In Phoenix there was a large increase in park visitation seven months after the SAHO.

By distinguishing urban from urban-fringe parks, we gained insights into on how park location influenced visitation trends. We hypothesized that core urban parks would exhibit a higher visitation trend as compared to urban-fringe parks during the period when residents were instructed to stay close to home. Figure 3 shows that there are similar increasing trends in urban-fringe park usage compared to pre-SAHO for both Chicago and Los Angeles. However, in Chicago, there was an overall decrease in all park usage (urban and urban-fringe). In Los Angeles we saw a steady monthly increase in urban-fringe park visitation following the SAHO, reaching pre-SAHO rates three months after the SAHO was enacted. Urban park visitation also increased, however it did not reach pre-SAHO levels.

Addressing another dimension of the socio-geographical context, we grouped the parks by their SVI and analysed the percentage change in visitation for those parks in the first interquartile range (IQR) and the last IQR of SVI. We found that across all cities and

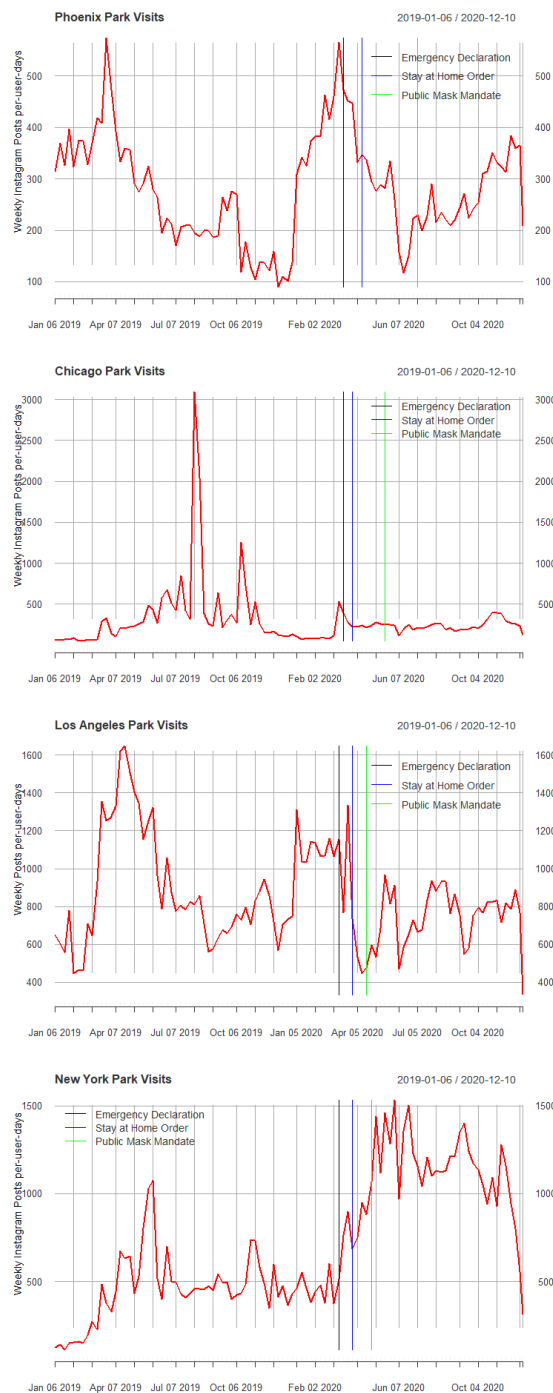


Figure 1: Time series of weekly PUD from 2019–2020, in four US cities. Major COVID-related policies are denoted as vertical lines.

months, parks that were located in poorer neighborhoods exhibited a significantly larger decrease in visitation 5 and 7 months post SAHO (Welch Two Sample Test $p < 0.09$). Interestingly, this trend

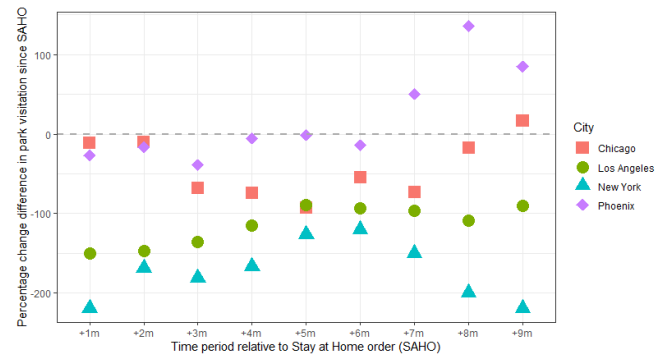


Figure 2: Percentage change of monthly park visitation from 2019 to 2020 relative to the SAHO enactment date. The dashed horizontal line represents the normalized average pre-COVID change in park visitation measured for each city.

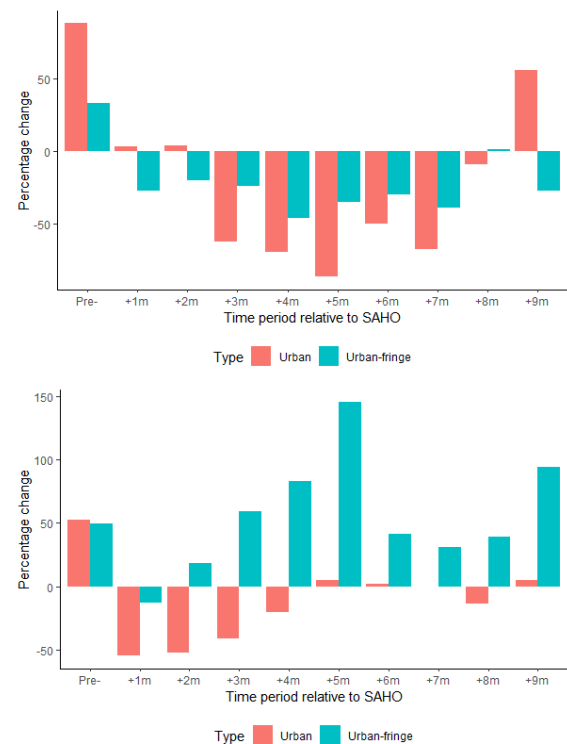


Figure 3: Percentage change of monthly park visitation from 2019 to 2020 in Chicago (left) and Los Angeles (right) relative to the local SAHO enactment date for urban and urban-fringe parks.

does not hold for pre-SAHO and earlier months of SAHO. Although these data are limited and do not capture the socio-economic level of all visitors, they do present a behavioral shift that would be worth further study.

4 DISCUSSION

This study is aimed at measuring and understanding changes in park visitation caused by the COVID-19 pandemic. Our survey of parks in four U.S. metropolitan areas shows that while there are shifts in park visitation coincident with the pandemic, the four focal cities do not exhibit a uniform response to the pandemic and associated mandates. Furthermore, trends in visitation are not consistently related to the setting of the parks that we studied, in comparisons of urban and urban-fringe parks located in neighborhoods that vary in socio-economic status.

Our analyses demonstrate the tremendous value in social media and mobility data. However, the study is limited to merely one source of social media from ten parks in four cities because of the effort required to collect such data at large scales. Consequently, the results are only suggestive, based on analyses using one simple proxy measure of visitation derived from social media on Instagram. Although there is empirical evidence that Instagram PUDs are useful for approximating park visitation [21, 23], there are many potential issues to consider. For instance, not all park visitors share their experiences on social media, and Instagram users are likely not representative of all park users [15]. These limitations likely introduce uncertainty that masks important geographic or socio-demographic differences in park use before and during the COVID-19 pandemic. Furthermore, we lack basic and important information about visitors, such as where they originate. Other aspects to consider for future work include investigating the change of behavior that may have occurred in usage of platforms such as Instagram due to the social stigma associated with breaking the SAHO.

The lack of volume and resolution of data to fully understand how COVID-19 impacted park visitation highlights the urgent need for researchers to advocate for data and collection methods that can provide more contextual information about the behavior of park visitors. In much the same way that technology companies, scientists, and local authorities worked together to create infrastructure for tracking and tracing the spread of the COVID19 virus, a unified effort could be mounted to enable the collection of anonymous data for understanding the impacts of the pandemic on access to parks and green-spaces. If, for instance, Google provided disaggregated mobility data (per park and per groups of visitors), then it would be possible to assess its biases and limitations, and then appropriately leverage it for analyses that could inform management decisions. Discussions about how social computing researchers can address the increasingly restricted access to social media and innovation of alternative methodologies for addressing this data gap should be a key focus of the CSCW community.

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